Reconciling the Debt: A Quantitative Match of Pre-1790 Certificates and Post-1790 Stock

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1 Introduction

This report tracks our key questions about individuals whose names appear in both the pre–1790 loan office certificates and the post–1790 federal stock records. We refer to these linked individuals as *matches*. Our goal is to understand who these matched holders are and how their holdings changed. Because the loan office certificates yield the highest match rate, we focus on that source first. We begin with population–level patterns, then turn to notable individuals.

1. What share of total value is held by matches?
2. How do average holdings compare: matches vs. non–matches?
3. How do matched holdings change from pre–1790 to post–1790?
4. Which states show the highest match rates?
5. Which occupations are most common among matches?
6. Which matches increased their holdings, and were they prominent individuals?

2 Summary of Findings

Definition. "Matched" refers to individuals whose pre–1790 loan office certificates could be linked to post–1790 federal stock records.

- **Concentration of value.** Matched individuals are **4.32%** of all holders yet account for **12.53%** of total face value.
- **Larger pre–1790 holdings.** Matched individuals held more in pre–1790 certificates on average (**$9,037**) than non–matched (**$2,845**).
- **Net decline overall.** Among matched individuals, aggregate holdings fall from **$7,069,910** (pre–1790) to **$1,955,405** (post–1790); median percent change is negative.

- **But not for everyone.** About **19%** of matched individuals increase their holdings between periods.
- **State contrast. Connecticut** contributes **34.65%** of matched holders despite only **14.12%** of all holders, indicating over–representation in the matched set.
- **Occupations.** Pre–1790, **Esquires (lawyers)** hold the largest share (**21.19%**); post–1790, **Merchants** hold the largest share (**22.65%**). Merchant and Merchant–Esquire categories show net increases; most other occupations decline.

See Section 4 for methods and detailed tables.

3 Limitations

1. To count the number of unique individuals, I counted the number of unique names. The issue is if two people with the same name live in the same state, they will be counted as one individual.
2. We used a basic matching process because of the pre-1790 datasets current limitations. If the same name appeared in the pre-1790 and post-1790 datasets, it was considered a match.
3. Small changes to the reconciliation process seem to have noticeable changes. I made a small change and re-reconciled. Afterwards, Connecticut moved ahead of Pennsylvania for number of matched individuals.

4 Analysis

4.1 Percent of Total Debt Held By Matches

1. Create `matched_status` column: `true` if name was matched to name in post-1790 dataset, `false` otherwise
2. Calculate the total face value of loan office certificates. This turned out to be $56,419,829.
3. Calculate the total owned by matches. This turned out to be $\$7,069,910$.

Approximately 12.53% of the loan office certificates were held by matches. To determine if this was a disproportionate share, we calculated the total number of unique individuals in the loan office certificate dataset. We then calculated the number of unique individuals that were matched.

$$\text{Percent} = \frac{\text{Total Unique Matched Names}}{\text{Total Unique Names}}$$

Matched individuals made up roughly 4.32% of the dataset. Therefore, matches held a disproportionate share of the total loan office certificates.

4.2 Average Amounts: Matched vs. Non-Matched

We will look at the average amount held by matches versus non-matches. Doing this builds off of the work done in [2.1]. We will see if matched individuals tended to hold more than non-matches. An issue that we have to account for is that one individual may have multiple rows of data. We have to combine these rows first. When grouping, we are assuming each unique name corresponds to one unique individual. Therefore, even if there may be two people named "John Smith" in Pennsylvania, we will assume them as one person. This is because we don't have a reliable way to tell them apart just yet. However, I doubt this would radically skew the data.

1. Filter for matched individuals.
2. Group rows by unique names, summing their certificates.
3. Calculate the average amount held by matched individuals.

Repeat the same process above for non-matched individuals. The results were the following:

- Matched: $\$9,037$ per person
- Non-Matched: $\$2,845$ per person

Therefore, matched individuals held on average $9,037 - 2,845 = \$6,192$ more than non-matched individuals. We won't make any definitive conclusions yet whether this indicates matched individuals were wealthier than non-matched individuals. We can make a stronger case for that once we look at geography, occupation, and external biographies.

4.3 Comparing Holdings Before and After 1790

We will see how much each matched individual owned before and after 1790. I used the `final_total_adj` column in the post-1790 dataset and compared it to the `Face Value` column in the pre-1790 dataset.

1. Create a unique id for each matched individual in the pre-1790 dataset corresponding to their row in the post-1790 dataset.
2. Iterate through each matched individual in the pre-1790 dataset, using their id to find their row in the post-1790 dataset.
3. Sum each individual's pre-1790 certificates and find their total amount in the post-1790 dataset.

Here are the results:

- Pre-1790: `$7,069,910.00`
- Post-1790: `$1,955,405.04`

I also calculated the percent change between the pre-1790 and post-1790 datasets per matched individual. The median was -58.93% per individual. Therefore, most individuals reduced their holdings. I went ahead and plotted a histogram for matched individuals' percent changes. As expected, most individuals had a negative percent change. However, there were notable outliers. One individual increased their holdings by $> 7000\%$ and 108 individuals at least doubled their holdings. I will look into these individuals in greater depth in a later section.

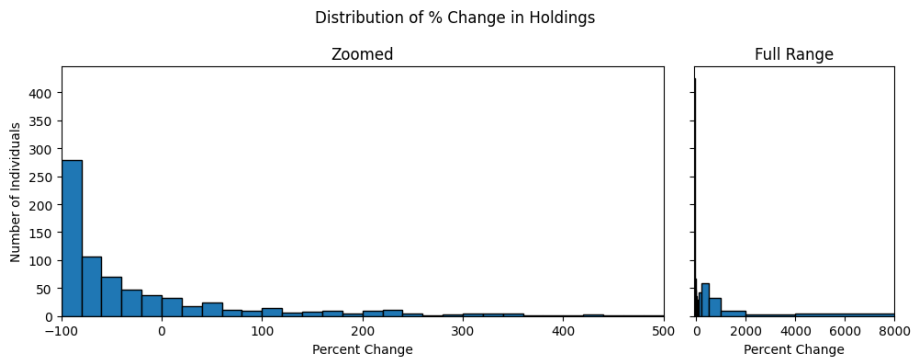

Figure 1: Distribution of Percent Changes in Holdings

4.4 Geographical Analysis

I want to find out which states had the most matches. Using Pandas `groupby` and `sort_values` methods, I sorted states by their number of matched individuals.

State	Unique Individuals	Share of All Matches (%)	Share of All Holders (%)
CT	271	34.65	14.12
PA	261	33.38	38.81
MD	86	11.00	5.57
NY	72	9.21	8.04
NH	46	5.88	2.93
MA	31	3.96	15.95
NJ	12	1.53	10.35
VA	2	0.26	3.17
DE	1	0.13	1.05

Table 1: Debt holders and their share of matched vs. total debt holders by state

To count the number of debt holders we calculated the number of **unique** names in the data. Connecticut has the most people matched with 271 debt holders found in the loan office certificate data set and the post-1790 data set. It has roughly 34.7% of all matched holders. This is surprising, considering that Pennsylvania had the most records in the loan office certificates dataset. The `% of All Debt Holders` column is the total number of individuals in each state divided by the total number of individuals in the loan office certificates. Pennsylvania had roughly 33.38% of matched individuals and 38.81% of all the individuals in the loan office certificates. Connecticut had a similar total of matched individuals to Pennsylvania (271 vs. 261) despite holding a considerable smaller share of all debt holders (14.12%). There are two hypothetical explanations:

1. Connecticut debt holders were more likely to hold onto their certificates than Pennsylvania.

2. Connecticut may also have had better preserved data than Pennsylvania.

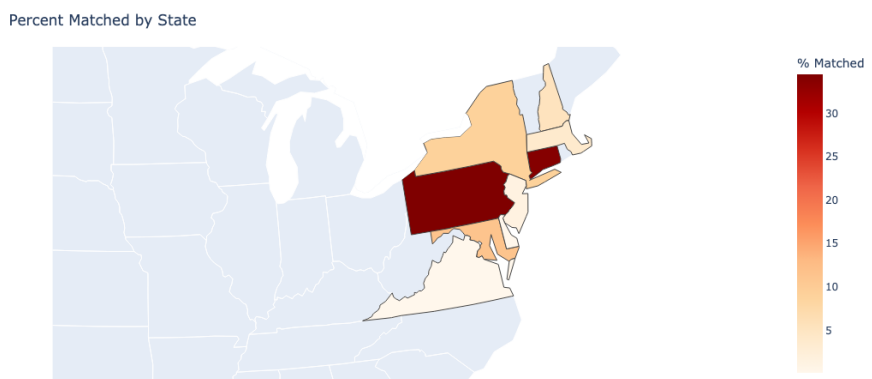

Figure 2: Percent Matched by State

4.5 Most Common Occupations

I want to find out what the most common occupations among matched individuals was. The purpose is to see whether certain occupations were more likely to hold onto debt certificates than others.

- Find the most common occupation for matched individuals using the post-1790 dataset.

- Find the most common occupations across the entire post-1790 dataset to compare against.

- Find how much each occupation owned with regards to pre-1790 and post-1790 securities.

Occupation	Count	Pre-1790 Total	Post-1790 Total	% of Pre-1790 Total	% of Post-1790 Total	Net % Change
Farmer	80	109300	45153.13	8.95	4.14	-58.69
Esquire	75	258800	194783.06	21.19	17.88	-24.74
Merchant	48	202800	246850.69	16.61	22.65	21.72
Widow	8	5200	4774.41	0.43	0.44	-8.18
Esquire — Merchant	6	45700	65757.27	3.74	6.03	43.89

Table 2: Summary of counts and totals by occupation, pre- vs. post-1790

I decided to only include the top five most common occupations. Few individuals held the other occupations. There were interesting points to note. Farmer was the most common occupation among matched individuals but they held a small amount of the debt relative to merchants and esquires. The net change in pre-1790 and post-1790 holdings decreased for every occupation except merchants and esquire-merchants. Merchants had a net increase of 21.72%. Meanwhile farmers' holdings decreased with a net change of (-58.69%). This begs the question: Did merchants purchase these certificates from farmers?

4.6 Who Increased Their Holdings?

This section is a continuation from [2.3]. Although most matched individuals' holdings decreased, a few individuals increased their holdings. We will look at the top 10 individuals whose holdings increased. We will see if they are notable individuals and if they appear in any historical records.

#	Recon ID	Name	State	Pre-1790	Post-1790	% Change	Occupation	Town
549	A311:Z311	Benjamin Harwood	MD	1500.00	645029.74	42901.98	—	Annapolis
368	A2297:Z2297	Joshua Lathrop	CT	300.00	24607.93	8102.64	—	Norwich
332	A2176:Z2176	Joseph Boggs	PA	200.00	15628.78	7714.39	Broker	Philadelphia
505	A2957:Z2957	Robert Buchanan	PA	600.00	38453.05	6308.84	Merchant	Philadelphia
40	A1192:Z1192	Henry Hill	PA	500.00	25583.82	5016.76	Administrator, Esquire	Philadelphia
71	A1319:Z1319	Isaac Wikoff	NJ	200.00	9094.72	4447.36	Esquire	—
131	A1561:Z1561	James Williams	MD	1300.00	37942.33	2818.64	—	Annapolis
482	A2832:Z2832	Philip Schuyler	NY	1600.00	34783.82	2073.99	—	Albany Ward 1
230	A1871:Z1871	John Keble	PA	200.00	4219.00	2009.50	Excise Officer	Philadelphia
471	A279:Z279	Barnabas Deane	CT	400.00	7109.75	1677.44	Merchant	Hartford

Table 3: Reconciled individuals with pre- and post-1790 totals, occupations, and towns

After looking at these individuals a few patterns emerge.

1. Founding fathers regularly show up in these individuals' biographies. These individuals interacted with well-known early Americans including George Washington, Thomas Jefferson, Benjamin Franklin, and Alexander Hamilton. America was a much smaller nation at the time so colonial elites probably interacted with one another much more often.

2. Many of them served in the military (Benjamin Harwood, Henry Hill, Philip Schuyler). It's not a coincidence they also purchased loan certificates. My guess is they viewed it as a form of patriotism.

3. Many individuals were merchants including Benjamin Harwood, Henry Hill, and Barnabas Deane. In order to buy securities, you needed money.

Benjamin Hardwood: Well-known merchant from Annapolis, Maryland. He established an import business with his brother, Thomas. He also served as a militia lieutenant. After the war, he was appointed Commissioner of Loans in 1777. This appointment makes me believe that most of his subscriptions were not owned by him directly. He also lent a lot of money. He had $88,000 outstanding when he died. Therefore, he must have had great personal wealth himself. He was also well-respected as per his obituary in the Maryland Gazette.

Joshua Lathrop: He owned a drugstore with his brother, Daniel. He graduated from Yale. He was a well-respected doctor.

Henry Hill: A prominent merchant. He made most of his money from the Madeira wine trade. He knew George Washington during his time serving as a colonel in the military. George Washington even stayed at his home when visiting Pennsylvania. He provided supplies for the Continental Army. He became a director for the Bank of North America after the war.

James Williams: Mayor of Annapolis (1793-1794)

Philip Schuyler: One of the most interesting debt holders. He was a US Senator from New York after the war. He was the father-in-law of Alexander Hamilton. His daughter, Elizabeth, married Alexander Hamilton. He even has a non-speaking role in the musical. He served in the US military as a Major General. He was a delegate to the Second Continental Congress. He was a Federalist and close ally of his son-in-law. His family was already prominent in New York and he made a lot of money from land speculation.

Barnabas Deane: He was a merchant who handled shipments from France. He wrote to Benjamin Franklin who was in Paris at the time. His brother, Silas Deane, is more well-known. Silas was an envoy to France and a delegate to the Constitutional Convention. Silas would later be disgraced for mismanaging his finances.